Linzhe Jiang

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Education

University College London, London, UK Sep 2024 – Sep 2025

MSc in Medical Robotics and Artificial Intelligence — Distinction (71.86/100)

Southwest Jiaotong University, Chengdu, China Sep 2019 – Jul 2023

Joint BEng in Mechanical Design, Manufacturing and Automation — 82.87/100

University of Leeds, Leeds, UK Sep 2019 – Jul 2023

Joint BEng in Mechanical Engineering — 82.87/100

Research Interests

Surgical Computer Vision; Surgical Spatial Intelligence; 3D Reconstruction and Registration; Vision-Language Models; Image-Guided Surgical Navigation

Publications

Peer-Reviewed Publications

[1] Huang, J., He, R., Khan, D. Z., Mazomenos, E. B., Stoyanov, D., Marcus, H., **Jiang, L.**, Clarkson, M. J., & Hoque, M. I. (2026). Surgical AI Copilot: Energy-based Fourier gradient low-rank adaptation for surgical LLM agent reasoning and planning. *Proceedings of the AAAI Conference on Artificial Intelligence, 40*(3), 1846–1854. doi:10.1609/aaai.v40i3.37164

[2] Wang, F., Liao, W., Su, H., Xu, L., Xue, H., **Jiang, L.**, Yang, Y., Piao, M., Sun, J., & Li, T. (2026). A histopathologically verified dataset of magnifying narrow-band imaging endoscopy for classifying the gastric precancerous cascade. *Scientific Data*. Advance online publication. doi:10.1038/s41597-026-07567-8

[3] Li, T., Liu, P., **Jiang, L.**, Gao, Y., & Wu, R. (2024). Four hours duration acts as the safety threshold for driving fatigue management. In *2024 IEEE International Conference on Computational Intelligence and Virtual Environments for Measurement Systems and Applications (CIVEMSA)* (pp. 1–6). doi:10.1109/CIVEMSA58715.2024.10586580

[4] Liu, X., **Jiang, L.**, Wang, H., & Jiang, C. (2020). Importance of effective regulation for the Chinese genetic testing industry. *Trends in Biotechnology, 38*(6), 577–579. doi:10.1016/j.tibtech.2020.01.005

Workshop Papers

[5] Liu, S., Bao, C., Cui, Z., Chu, X., Ren, B., Gu, L., . . . , **Jiang, L.**, . . . , & Huang, C. (2026). NTIRE 2026 3D restoration and reconstruction in real-world adverse conditions: RealX3D challenge results. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)* (pp. 2119–2136). arXiv:2604.04135

[6] **Jiang, L.**, Huang, J., Zhang, C., Jiang, C., Mao, Z., & Hoque, M. I. (2026). RefineRank: Joint box refinement and ranking for surgical spatio-temporal grounding. In *Proceedings of the ECCV 2026 MedVidU Workshop* (non-archival). **Ranked 1st in STG score** on the MedVidBench leaderboard. arXiv:2608.23928

Preprints & Manuscripts in Preparation

[7] **Jiang, L.**, Huang, J., Bano, S., Clarkson, M. J., Mao, Z., & Hoque, M. I. (2026). C3VDReg: A benchmark for local-to-local colonoscopic registration toward anatomical localization. *arXiv*. arXiv:2511.00260.

[8] **Jiang, L.** (2026). VGGT-Stream: Sparse-depth verified memory admission for long-horizon feed-forward 3D reconstruction. Manuscript in preparation.

[9] **Jiang, L.** (2026). From value constraints to anatomical-prior-guided dynamic spatial state: A perspective on hardware-agnostic surgical navigation. *Perspective article in preparation*

Research Experience

Tianjin Union Medical Center, The First Affiliated Hospital of Nankai University, Tianjin, China Oct 2025 – Present

Visiting Student Researcher, Department of Thoracic Surgery
Supervisor: Chunyang Jiang, M.D., Chief Physician

- Observed 10 complete thoracoscopic wedge resections and 3 hybrid electromagnetic-navigation procedures, identifying intraoperative lung deformation as the central barrier to preoperative-image-based navigation
- Designed a DICOM-to-3D reconstruction workflow for two patients undergoing lung nodule resection: AGU-Net airway segmentation, HiPaS pulmonary artery–vein separation, MONAI/MedSAM2 nodule segmentation, with agent-orchestrated topological QC and nnInteractive repair
- Generated labeled, Taubin-smoothed meshes and color-coded 3D-printed models used by a thoracic surgeon for preoperative planning; postoperative review of both cases confirmed agreement with intraoperative findings

The University of Manchester, Manchester, UK Aug 2025 – Present

Research Assistant, School of Health Sciences (Remote)
Supervisor: Mobarak I. Hoque, Associate Professor

- Curated 50 paired MRI-video pituitary-surgery cases from 200+ clinical records and built an ITK-SNAP-based MRI-to-point-cloud pipeline to extract sellar anatomical references for downstream 3D reconstruction and registration.
- Reconstructed intraoperative video of the sellar region with VGGT Omega, a feed-forward 3D reconstruction model that outperformed Depth Anything V3 and an in-house trained depth model, though relative scale remained inaccurate and the region lacked reliable visual landmarks
- Manually registered MRI-derived static point clouds to video-derived dynamic point clouds and recovered metric scale from anatomical inter-landmark distances; surgeon review found reliable anatomical correspondence in only 3 of 50 cases, revealing the limitations of real-world MRI-video pairs for learned registration and motivating the development of PitvisGym with controlled, registered point-cloud pairs.
- Developing PitvisGym, an MRI-constrained simulator prototype built from one high-quality real case; it supports virtual endoscope navigation inside the nasal cavity, renders simulated endoscopic views, and displays the endoscope pose in MRI in real time.

Peking Union Medical College, Beijing, China Sep 2023 – Jul 2024

Research Assistant, Institute of Biomedical Engineering
Supervisor: Ting Li, Professor, Chinese Academy of Medical Sciences & Peking Union Medical College

- Simulated photon transport for an fNIRS-based clot detector using MCVM on 3D voxelized lower-limb models 1×10^7, evaluating the effects of tissue absorption, scattering, and anisotropy on photon propagation and demonstrating penetration beyond 4.2 cm under the modeled configuration.
- Fabricated and assembled a prototype fNIRS clot detector through circuit-board soldering and flexible probe construction, contributing to the CICSIC 2024 Silver Award project (non-invasive circulatory monitoring for elderly care); system tests showed channel crosstalk below the noise floor, and cuff-occlusion experiments tracked expected hemoglobin changes
- Conducted fNIRS-based driver-fatigue experiments (8 volunteers, 7-hour simulated driving, hourly visual-involuntary-attention tests); the CIVEMSA 2024 study identified a behavioral fast phase beginning around 4 hours, supporting a 4-hour continuous-driving safety threshold

Selected Projects

Surgical Spatio-Temporal Grounding (RefineRank)

- Built a compact 1.25M-parameter RefineNet coupling frozen MedVLM and GroundingDINO to refine detector coordinates and box-quality scores without end-to-end backbone alignment.
- Ranked 1st in STG score on MedVidBench and improved mIoU from 0.2719 to 0.4534 (+66.8%); oracle analysis identified detector-recall and stale-feature limitations, while thoracic-surgeon review suggested value for education and postoperative analysis.

Cross-Modal Point-Cloud Registration in Surgery

- Developed MambaNetLK by replacing PointNetLK's PointNet feature extractor with a Mamba SSM for colonoscopic point-cloud registration, achieving the best performance among PointNetLK-family methods on C3VDReg.
- Built C3VDReg, a public benchmark derived from C3VD with 10,015 viewpoint-matched colonoscopic point-cloud pairs under a fixed evaluation contract; identified translation ambiguity in repetitive tubular anatomy as the dominant failure mode

Long-Horizon Feed-Forward 3D Reconstruction (ongoing)

- Developing a sparse-depth, reprojection-verified memory mechanism for VGGT-style long-horizon 3D reconstruction to reduce drift and uncontrolled memory growth.

Undergraduate Thesis: Path Planning of a Six-Axis Manipulator

- Independently re-derived and validated UR5 closed-form inverse kinematics (8 analytical solutions) for end-effector-to-joint-space trajectory conversion.
- Improved the artificial potential field method with local-minimum escape and re-entry avoidance; evaluated it against four APF variants in three self-built MATLAB 3D scenes.

Honors & Awards

- Silver Award, China International College Students' Innovation Competition (CICSIC), 2024 — non-invasive circulatory monitoring for elderly care; contributed detector hardware and optical-parameter study
- Organizing Committee Member, 4th Plenary Meeting of ISO/IEC JTC 1/SC 43 (Brain–Computer Interfaces), 2024

Patents

[T1] Support Fixation Bracket for Endoscopic Surgery. CN Patent No. ZL 2022 2 2217588.9

[T2] Percutaneous Pulmonary Puncture Angle Assist Device. CN Patent No. ZL 2023 2 2773803.8

Technical Skills

Programming: Python, C, C++, C# — *Python in all projects; C/C++/C# for coursework/embedded firmware*

Deep Learning & Foundation Models: PyTorch, Hugging Face Transformers, Accelerate, DeepSpeed, vLLM, LLaMA Factory; Vision Transformers (ViT), Mamba, Qwen/Qwen-VL, CycleGAN — *RefineRank, MambaNetLK, Surgical AI Copilot*

Computer Vision & 3D: COLMAP, SLAM, NeRF, 3D Gaussian Splatting, feed-forward reconstruction (VGGT, Depth Anything), point-cloud registration — *C3VDReg, VGGT-Stream, Manchester/Tianjin reconstruction*

Robotics & Engineering: ROS/ROS 2, MATLAB Robotics Toolbox, SolidWorks, Abaqus, ITK-SNAP, 3D Slicer, CloudCompare, MCVM — *undergraduate thesis, fNIRS prototype, medical image segmentation; SolidWorks/Abaqus for undergraduate mechanical design and FEA*